

From Black Box to Glass-Box Brain: Bridging Sensorimotor Systems and Circuits via Mesoscale AI Link

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Current neuroimaging methods (functional magnetic resonance imaging, fMRI and magnetoencephalography, MEG) accurately map **where** and **when** brain activity occurs answering associative neuroscience research questions but lack mechanistic insight into **how**^{1,2}. The Glass-Box Brain (GBB) project aims to address the black-box problem of neuroimaging by introducing a transparent and interpretable artificial neural network architecture trained with novel 7-Tesla mesoscale, fMRI data of the human sensorimotor system. This model uses indirect neuroimaging data to solve multiple inverse problems to yield biologically relevant parameter maps that can be validated with separate experiments. Answering the **hows** of neuroimaging is a step toward mechanistic neuroimaging from conventional associative and descriptive neuroimaging (where and when). Since the human brain is computationally incredibly an energy-efficient organ when compared to current machine learning architectures³, the GBB architecture could also inform us of more efficient novel machine learning design choices regarding architectures and learning algorithms.

The GBB architecture consists of three distinct components (see Figure 1). The first component is a feature extractor layer that has six alternative neural network architectures. These alternative architectures correspond to input to a neural population (afference) and function as alternative hypotheses to be tested to examine which spatial and temporal scales are most relevant for information processing in the sensorimotor cortices. The second component corresponds to the internal neural computations and information processing of a neural population and consists of closed-form continuous-time (CfC) neural networks⁴, which model the internal computations of a neural population with interpretable differential equations. The third component corresponds to connectivity between different neural populations and consists of a variant of Kolmogorov-Arnold networks (FastKAN)⁵ and model these connections with a mixture of Gaussian basis functions. The FastKAN component can also contain several attention-heads that can model different types of connectivity such as excitatory, inhibitory and neuromodulatory connectivity or even different types of neurotransmitter systems (e.g. GABAergic, dopaminergic). Both CfC and FastKAN layers are interpretable and can be used to extract different spatial parameter maps projected on a group-brain after training with the fMRI data.

The GBB architecture functions as a computational analogue of the human sensorimotor system that can be used in crude/simplified simulations of its function after training as well as for extracting different mechanistic properties of different brain regions using the spatial parameter maps projected onto a brain template. In this sense, the GBB architecture functions as a computational microscope that can generate novel, testable hypotheses that are not ground-truths before further experimental validation. Hyperparameters can also be used to generate new hypotheses about the sensorimotor system and aid in designing novel machine learning architectures.

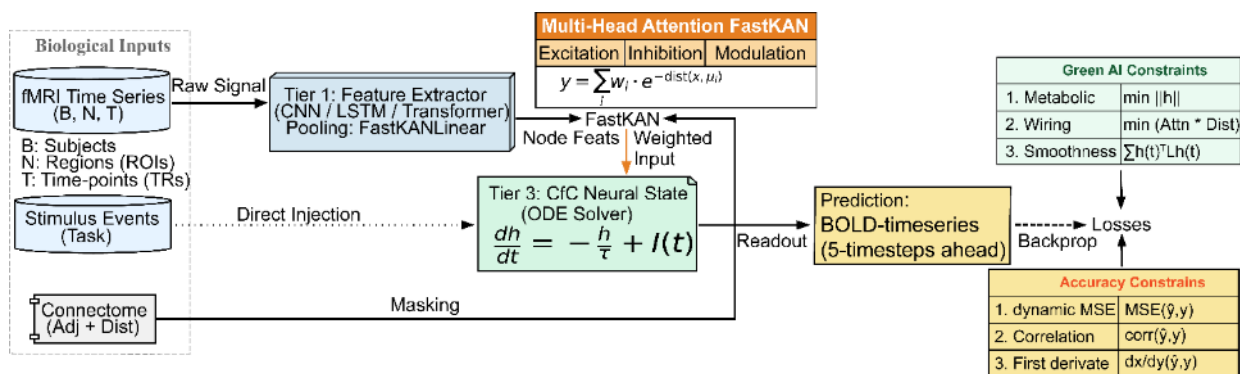


Figure 1. **Flowchart of the GBB architecture and its function.** This figure shows the basic architecture of the GBB in a flowchart.

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